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# Anomaly Transformer for Vibration Data Analysis: Case Study on CWRU Dataset

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***Abstract*—Time series anomaly detection has a wide range of applications in various fields of society, one of which is the detection of anomalies in vibration data. Abnormal vibration data often indicates the existence of potential disasters or abnormal operation of equipment, so timely detection and identification is important for the prevention of earthquakes and improvement of industrial production efficiency. In this paper, an unsupervised time series anomaly detection algorithm, Anomaly Transformer, is used to detect anomalies in CWRU vibration datasets.Anomaly Transformer leverages the powerful characterization capabilities of the Transformer network and an improved anomaly attention mechanism to capture anomalies in the time series. Experimental results show that Anomaly Transformer exhibits excellent performance in the CWRU vibration dataset, achieving high precision and recall. In addition, Anomaly Transformer demonstrated a low false alarm rate, indicating its high accuracy in recognizing real anomalies. Future research can further explore the appli- cation of Anomaly Transformer in other domains with deeper optimization and extensions.**

***Keywords-time series anomaly detection; deep learning; vibration data***

## I. Introduction

With the development of various fields of society, there is an increasing demand for vibration data anomaly detection. Vibration data, as a kind of common time series data, is widely used in fields such as industrial production, structural monitoring and geological exploration. By analyzing and detecting anomalies in vibration data, it can help to quickly identify potential failures, predict the health of structures, or determine the occurrence of geologic hazards. However, accurately detecting and recognizing anomalies in vibration data has been a challenging problem due to the high dimensionality, complexity and noise interference of vibration data.

Since the data explosion and the rise of deep learning in the early 21st century, temporal data anomaly detection algorithms have gained rapid development and widespread attention. Over the past decades, temporal data anomaly detection algorithms have gone through several development stages, each of which introduced classical algorithms to address different challenges. Traditional timing data anomaly detection algorithms based on statistical methods were among the earliest methods ap- plied. These methods include algorithms based on probabilistic statistics, such as the Z-score method [1] based on mean and standard deviation. They identify anomalous data points that are significantly different from normal patterns by statistically analyzing the data. These methods usually assume that the data obeys a particular probability distribution, but vibration data contains complex nonlinear and non-Gaussian distributed patterns, and thus may not accurately characterize the true distribution of vibration data. Subsequently, machine learning-based anomaly detection algorithms for time-series data have come to the fore. These algorithms use machine learning models to model and train the data to distinguish between normal and abnormal patterns. Classical machine learning algorithms include distance-based algorithms, clustering-based algorithms, and classification-based algorithms. However, traditional machine learning algorithms require manual design and extraction of features. For vibrating data, selecting appropriate features and representing them effectively is a challenging task. With the rise of deep learning, neural network-based anomaly detection algorithms for timing data have made significant progress. Among them, convolutional neural networks (CNNs) [2] are widely used to process time series data, utilizing their hierarchical feature extraction capabilities to capture anomalous patterns in the data. In addition, Recurrent Neural Networks (RNN) [3] and its variants, such as Long Short-Term Memory Networks (LSTMs) [4] and Gated Recurrent Units (GRUs) [5],



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are capable of modeling time series data and detecting anomalies by learning temporal dependencies of the data. However, their lack of interpretability in data anomaly detection as well as the possibility of experiencing gradient vanishing or gradient explosion when dealing with long-term dependencies make it difficult to capture long-term temporal relationships. Recently, emerging techniques such as Attention Mechanism (AM) [6] and Generative Adversarial Network (GAN) [7] have also been introduced to temporal data anomaly detection. Meanwhile, with the excellent performance of Transformer [8] in fields such as NLP, more and more researchers have started to apply it to the field of temporal data anomaly detection. Its strong modeling capability and ability to capture long-term dependencies can well solve the complex temporal dependencies and long-term temporal patterns existing in time- series data.Its parallel computing capability also makes it more efficient in dealing with large-scale vibration data.

In summary, we employ an unsupervised time series anomaly detection algorithm, Anomaly Transformer [9], to detect anomalies in vibration data. Different from Transformer's self-attention mechanism, Anomaly Transformer [9] proposes a new attention mechanism, Anomaly-Attention, which contains two branching structures for modeling a priori associations and sequence associations in a unified way, which in turn facilitates the computation of association differences. The a priori association uses a learnable Gaussian kernel function that exploits the single-peak feature of the Gaussian distribution to make the data more attentive to neighboring points, while the sequence association is obtained from the attention computation in the standard Transformer [8]. Then a very large and very small strategy is used to amplify the gap between normal and abnormal points, and finally a new judgment criterion for anomaly detection is formed.Anomaly Transformer [9] achieves excellent results on the CWRU vibration dataset.

## II. Related works

Timing anomaly detection is an active research area and many methods and techniques have been proposed for detecting anomalies in time series data. According to their models and techniques, time-series anomaly detection algorithms are usually categorized into two groups: traditional methods and deep learning methods.

Traditional methods are the early methods in the field of timing anomaly monitoring. Before the emergence of deep learning methods, traditional timing anomaly detection methods are mainly based on statistics and machine learn- ing techniques, and the basic idea is to identify and detect anomalies in timing data through feature engineering and machine learning techniques. They are usually subdivided into: statistical methods, which are one of the simplest and common traditional methods. They are based on modeling and anomaly detection based on statistical properties of the timing data. Commonly used statistical methods include mean, standard deviation, variance, root-mean-square, etc.; distance- based algorithms such as k-nearest neighbor algorithm (KNN) [10] and Isolation Forest [11]; clustering-based algorithms such as k-means algorithm (K-means) [12] and Gaussian Mix- ture Models (GMMs) [13]; and classification-based algorithms such as Support Vector Machines (SVMs) [14] and Decision Trees [15].

With the rise of deep learning, neural network-based timing anomaly detection methods have gained widespread attention. Deep learning methods are able to automatically learn more abstract and higher-level feature representations from raw data through the construction and training of multi-layer neural networks to better capture anomalous patterns in the data. Among them, Convolutional Neural Networks (CNNs) [2], Recurrent Neural Networks (RNNs) [3], Long Short-Term Memory Networks (LSTMs) [4], and Autoencoder [16] are widely used for modeling and detecting anomalies in temporal data.CNNs [2] excel in dealing with localized features and translational invariance, while RNNs [3] and LSTMs [4] are suitable to capture temporal dependencies in sequences. Autoencoder [16] is able to learn the feature representation of normal data and detect anomalies by reconstructing the errors. In addition, some deep learning algorithms incorporate techniques such as Attention Mechanisms [6] and Genera- tive Adversarial Networks (GAN) [7] to further improve the performance of anomaly detection. The attention mechanism automatically focuses on the important parts of the data to improve the performance of anomaly detection. Generative Adversarial Networks provide a new perspective to detect anomalous data through the ability to generate real samples.

However, these traditional deep learning methods have some limitations in timing anomaly detection. Therefore, in this paper, we adopt a Transformer-based temporal anomaly detection algorithm: the Anomaly Transformer [9]. Unlike traditional window-based methods, the Anomaly Transformer [9] captures long-term dependencies in a time series through global modeling and anomaly-attention mechanisms and provides an efficient representation of anomalous samples, presenting a very large very small strategy to obtain a significant difference in the association.

## III. Method

Suppose a measurement system continuously monitors m variables, and the time series of observations over time is represented by a set of time points $\{x_1, x_2..., x_N\}$, where $x_t \in R^d$ denotes the observation at time t. The goal of the article is to use an unsupervised algorithm to determine whether the time point xt is anomalous or not in the absence of labeling information.

### A. Overall Architecture

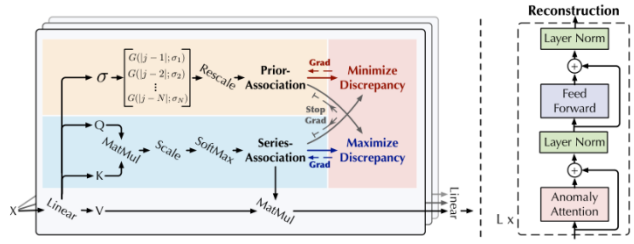


Figure 1. The overall architecture of the algorithm. On the left is the composition of Anomaly Attention, and on the right is the overall architecture. The picture cite from [9]

The overall architecture of Anomaly Transformer [9] is shown in Figure 1. Anomaly Transformer [9] is a structure consisting of alternately stacked Anomaly-Attention blocks and feed- forward layers. The design of this stacked structure enables



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the model to learn the underlying correlations and patterns from deep multi-layer features.The Anomaly-Attention blocks are able to adaptively capture key information in the time series through the anomaly-attention mechanism to effectively detect anomalies. By alternately stacking multiple Anomaly- Attention blocks, the model can progressively extract higher- level feature representations, giving it stronger modeling and prediction capabilities. The advantage of this stacking structure is that it can fully utilize the rich information of the deep features to improve the sensitivity and accuracy of the model to the anomaly patterns in complex time-series data. Different from Transformer's [8] self-attention, Anomaly Transformer [9] adopts the updated Anomaly attention, and at the same time employs the very large very small strategy to magnify the difference between anomalies and normal points, which is more obvious in the anomaly scores.

### B. Anomaly Attention

For time series data, each time point has a greater relationship with its neighboring points, and also has a greater association with other points on the sequence, the association with the neighboring points is called priority association, and the association with the sequence is called series association, which can also be understood that the neighboring points are the local information, and the series association is the global information. Therefore, Anomaly attention is divided into two branches, series association corresponds to the self- attention weights learned from the original sequence, and priority assoication is calculated based on the Gaussian kernel function, which utilizes the single-peak nature of the Gaussian kernel to make the results more focused on the relationship between adjacent time steps. The Gaussian kernel range is also adjusted by the introduced learnable scale parameter σ to accommodate the anomaly lengths of different time series patterns.

The association differences are quantified by formalizing the symmetric KL scatter between the prior association and the series association, and then averaging the association differences from multiple layers to generate a more informative metric, AssDis(P,S,X), which consists of associations from multiple levels of features [17].

### C. Minimax Strategy

To achieve the goal of unsupervised learning, we utilize the reconstruction error as a key metric for model optimization. The model's ability to characterize the data is evaluated by comparing the difference between the original data and the reconstructed data. As a general rule, the reconstruction loss should be equal to the loss of the minimum squared difference between the original and reconstructed data plus AssDis. this minimizes the loss while maximizing AssDis, making normal sequences more normal and increasing the distinction between anomalies and normals. However, directly operating in this way makes the scale parameter σ of the single-peak Gaussian kernel drastically smaller, leading to meaningless priority association. Therefore, a very large very small strategy is proposed to avoid this situation. In this strategy, two key phases are employed to achieve timing anomaly detection. First, in the minimization phase, the pri- ority association is used to approximate the learned series association in the original timing data.This process allows the priority association to adapt to different timing patterns and capture regular patterns in the data. Then, in the maximization phase, the differences between the different associations are maximized by optimizing the series association. This process allows series associations to focus more on non-proconsistent data points, which increases the difficulty of reconstructing anomalies. In this way it is possible to improve the detection of anomalies and provide more accurate anomaly detection results. Ultimately, the normalized association differences and reconstruction errors are combined to develop a new anomaly detection criterion.

## IV. Experiments

We performed model validation on the CWRU vibration dataset and the results show that the algorithm achieves excellent results on the CWRU dataset.

### A. Dateset

The CWRU bearing dataset is collected and maintained by the Department of Mechanical Engineering at Western Reserve University and is intended to provide a standardized benchmark dataset for evaluating and comparing the performance of different algorithms.The CWRU bearing dataset contains vibration data from rotating machinery systems, which includes vibration signals of bearings under different fault conditions. The dataset covers several fault types, such as inner ring faults, outer ring faults, rolling element faults, etc., with multiple instances of each fault type. These vibration data are recorded in the form of time series with high sampling rate and high resolution to provide detailed vibration characteristics. We selected normal data at 48 KHZ and rolling body failure data with a diameter of 7 mils as the data set for this test.

### B. Implementation details

When the bearing fails, the waveform graph shows that both the acceleration at the drive end and the acceleration at the fan end change significantly and are correlated, so the DE and FE data in the original CWRU dataset are fused to form a new dataset, and preprocessing, such as data cleaning, down-sampling, and normalization, is performed. A non-overlapping sliding window is used to obtain a set of subsequences, and the size of the sliding window is fixed at 20. Meanwhile, a threshold δ is set, and when the anomaly score at a certain time point is larger than the threshold, it is marked as anomaly. The distribution of anomaly labels is controlled by adjusting the threshold δ so that the proportion of data marked as anomalous in the validation dataset reaches a predetermined ratio r. We set r to 3. We use Adam for optimization, setting its initial learning rate to $10^{-4}$ , the batch size to 128, and the training period to 10 by default.

### C. Main Results

In this study, we used the Anomaly Transformer [9] algorithm for anomaly detection on the CWRU dataset and achieved satisfactory results. After experimental evaluation, the F1 score reaches 94.51, indicating that the algorithm achieves a better performance after integrating precision and recall. The results are shown in Figure 2. . It also shows that the model has high accuracy and reliability in detecting anomalies in the CWRU dataset. The recall rate of 98.92 implies that the algorithm is effective in capturing anomalies present in the CWRU dataset. The high recall indicates that the algorithm has a good ability to detect real anomalous data and avoid



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underreporting as much as possible. The precision is 90.47, which indicates that the algorithm is able to distinguish between normal and abnormal data effectively. The results are shown in Figure 4. . The high precision means that the algorithm is accurate in determining the normal samples in the CWRU dataset,reducing the possibility of false alarms. The Accuracy score of 95.94 represents the effectiveness and accuracy on this dataset. The results are shown in Figure 3. .

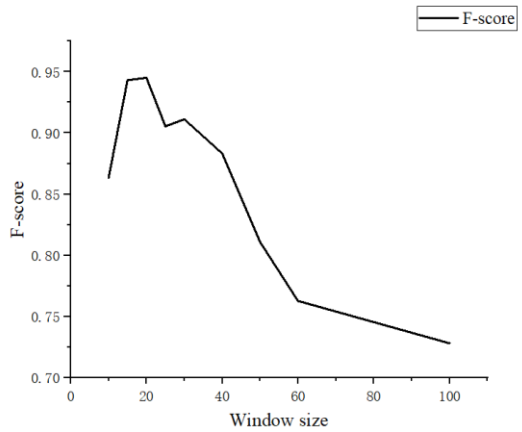


Figure 2. F-score score results under different window sizes when the anomaly ratio is fixed.

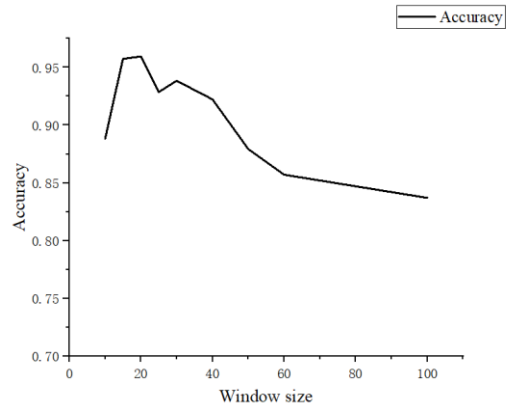


Figure 3. Accuracy score results under different window sizes when the anomaly ratio is fixed.

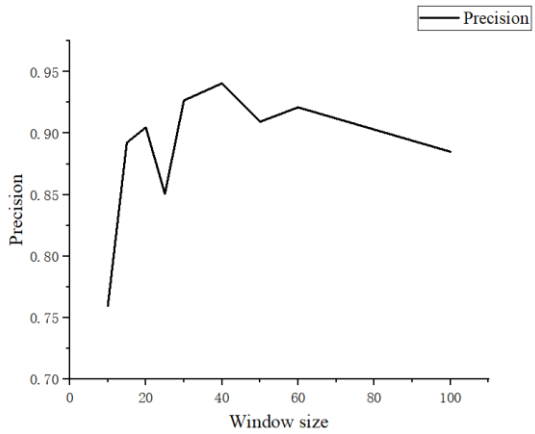


Figure 4. Precision score results for different window sizes when the anomaly ratio is fixed.

## V. Conclusion and future work

In this paper, we applied the Anomaly Transformer [9] algorithm for anomaly detection on vibration data.The algorithm is based on the Transformer [8] architecture with an updated anomaly attention mechanism, and therefore, it has the power to capture complex temporal dependencies and identify anomalous patterns in the data. Anomaly Transformer [9] has achieved excellent results on the CWRU vibration dataset, proving the accuracy and reliability of the algorithm in the field of vibration data anomaly detection. Vibration data plays an important role in geohazard monitoring, such as critical rock failures, landslides and the health of building structures. By applying the Anomaly Transformer [9] algorithm to the early warning monitoring of geohazard vibration data, we have reason to believe that the algorithm is also efficient and effective, and will provide strong support for geohazard risk management and mitigation in the future.

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